

Lee 17

①

- we saw that DT-LTI systems could be modeled as Difference equations (of I/P & O/P)
- and that these difference equations could be converted into simple algebraic (non-iterative) equations using Z-TX
- Particularly, using
$$x[n-k] \Leftrightarrow z^{-k} X(z) \quad \text{for zero ICs.}$$

→ e.g. a DT LTI system:

$$y[n] = 3x[n] + 0.8y[n-1]$$

→ Described by difference eqn. (contains delay samples etc.)

→ Using z with assumption ICs = 0 (for simplicity)

$$\Rightarrow Y(z) = 3X(z) + 0.8z^{-1}Y(z)$$

$$\Rightarrow Y(z) = \frac{3X(z)}{1-0.8z^{-1}}$$

$$Y(z) = H(z)X(z)$$

→ simple algebraic eqn. in z -domain.

→ on similar lines

(2)

① CT-LTI systems are represented by differential equations!

② And using Laplace Tx. these differential equations can be converted into simple algebraic equations.

③ we will make use of the fact that

$$\frac{d^k y(t)}{dt^k} \Leftrightarrow s^k Y(s) \quad \text{Assuming all } \mathcal{I}t_s = 0.$$

→ e.g.

Differential eqn. →

$$s \frac{dy(t)}{dt} + 6y(t) = 4x(t) \quad \mathcal{I}t_s = 0$$

$$\begin{aligned} \Rightarrow s s Y(s) + 6 Y(s) &= 4 X(s) \\ (s s + 6) Y(s) &= 4 X(s) \\ Y(s) &= \frac{4}{s s + 6} X(s) \end{aligned}$$

$$Y(s) = H(s) X(s)$$

→ simple algebraic eqn.

→ In general, an CT-LTI system described by (3)
Nth order DE (assuming zero ICs for convenience).

$$\frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + \dots + a_{N-1} \frac{dy(t)}{dt} \\ = b_0 \frac{d^N x(t)}{dt^N} + b_1 \frac{d^{N-1} x(t)}{dt^{N-1}} + \dots + b_N x(t)$$

→ This can be simplified using Laplace as

$$s^N Y(s) + a_1 s^{N-1} Y(s) + \dots + a_{N-1} Y(s) = b_0 s^N X(s) + b_1 s^{N-1} X(s) + \dots + b_N X(s) \\ \Rightarrow Y(s) (s^N + a_1 s^{N-1} + \dots + a_{N-1}) = X(s) (b_0 s^N + b_1 s^{N-1} + \dots + b_N)$$

$$Y(s) = \frac{b_0 s^N + b_1 s^{N-1} + \dots + b_N}{s^N + a_1 s^{N-1} + \dots + a_{N-1}} X(s)$$

$$Y(s) = H(s) X(s)$$

→ $H(s)$ = System Transfer function.

→ Also Zero-state Response
(since we assumed ICs = 0).

→ checking stability of CT-LTI from $H(s)$ (4)

→ BIBO stable from $h(t)$ if

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

→ what about BIBO stability from $H(s)$?

→ of zeros & poles

Zeros: values of s that make $H(s) = 0$

Poles: values of s that make $H(s) = \infty$

→ For rational case (i.e. LTI case) $H(s)$ can be written as rational function e.g.

$$H(s) = \frac{P(s)}{Q(s)} = \frac{(s-b_1)(s-b_2)\dots(s-b_n)}{(s-a_1)(s-a_2)\dots(s-a_m)}$$

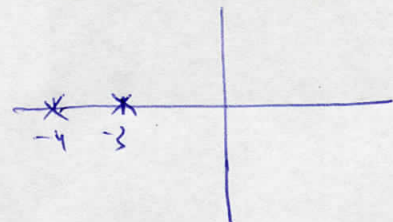
$\Rightarrow H(s) = 0$ for $s = b_1, b_2, \dots, b_n$

$H(s) = \infty$ for $s = a_1, a_2, \dots, a_m$

→ Case: all poles in left half plane

$$H(s) = \frac{1}{s+3} + \frac{2}{s+4} \Rightarrow h(t) = \underbrace{e^{-3t}u(t) + 2e^{-4t}u(t)}_{\text{both decaying functions.}}$$

$$\Rightarrow \int_{-\infty}^{\infty} |h(t)| dt < \infty$$



(5)

if all poles in left half plane (as they lead to decaying functions in time domain)

→ case: even a single pole in RHP

$$H(s) = \frac{1}{s+3} + \frac{2}{s-4}$$

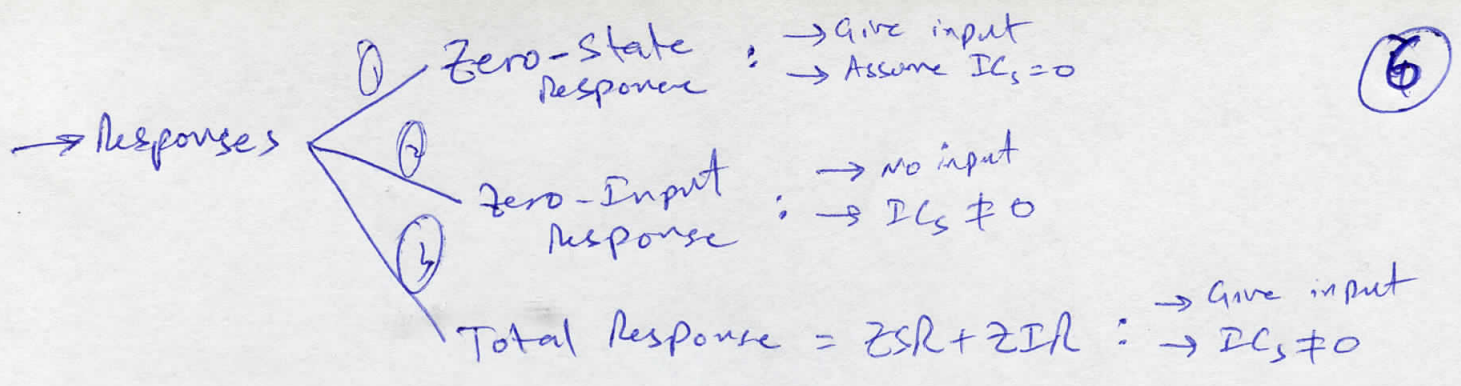
$$\Rightarrow h(t) = \underbrace{e^{-3t} u(t)}_{\text{decaying}} + \underbrace{2e^{4t} u(t)}_{\text{exponentially growing}}$$

now

$$\int_{-\infty}^{\infty} |h(t)| dt = \infty$$

∴ CT-LTI BIBO stable iff all poles of $H(s)$ in LHP!

↓
Assuming $H(s)$ in reduced form
(common pole and zero cancelled)



→ example: Consider CT LTI system.

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

① let's do ZSR i.e. $IC_s = 0$

→ some input! let's say $x(t) = e^{-4t} u(t)$
 $\Rightarrow X(s) = \frac{1}{s+4}$ (table)

→ Find $y(t)$!

→ note: easier if we use Laplace

→ using Laplace table with zero IC_s gives

$$s^2 Y(s) + 5s Y(s) + 6Y(s) = sX(s) + X(s)$$

$$\Rightarrow Y(s) = \frac{s+1}{s^2+5s+6} X(s) = \frac{s+1}{(s^2+5s+6)(s+4)}$$

→ using PFE $Y(s) = \frac{-\frac{1}{2}}{s+2} + \frac{2}{s+3} - \frac{3/2}{s+4}$

→ From table of $\mathcal{L}$

$$y(t) = \left(\frac{1}{2} e^{-2t} + 2e^{-3t} - \frac{3}{2} e^{-4t} \right) u(t)$$

→ ZSR

② now let's do Zero-Input Response.

$$\rightarrow x(t) = 0 \quad \forall t$$

$$\rightarrow \text{some ICs!} \quad \text{eg. } y(0) = 2, \quad \dot{y}(0) = 1$$

$\rightarrow$ same system, but now

$$\frac{dy(t)}{dt} \Leftrightarrow sY(s) - y(0)$$

etc.

no E/P

$\downarrow$

$$\Rightarrow \underbrace{\left[s^2 Y(s) - 2s - 1 \right]}_{\frac{d^2 y(t)}{dt^2}} + \underbrace{s \left[s Y(s) - 2 \right]}_{\frac{dy(t)}{dt}} + \underbrace{6 Y(s)}_{y(t)} = 0$$

from
ILs
(not
E/P)

$$\Rightarrow s^2 Y(s) + 5s Y(s) + 6 Y(s) = 2s + 11$$

$$\Rightarrow Y(s) = \frac{2s + 11}{s^2 + 5s + 6} = \frac{7}{s+2} - \frac{5}{s+3}$$

$$\rightarrow \text{using Laplace Table: } y(t) = (7e^{-2t} - 5e^{-3t})u(t)$$

$\downarrow$
ZIR

③ The case of Total Response i.e.

$$\rightarrow ICs \neq 0$$

$$\rightarrow \text{Input} \neq 0$$

$\rightarrow$ for example above, you can verify that

$$TR = ZIR + ZSR$$

$$\Rightarrow y(t) = (7e^{-2t} - 5e^{-3t})u(t) + \left(-\frac{1}{2}e^{-2t} + 2e^{-3t} + \frac{3}{2}e^{-7t}\right)u(t)$$